

Es Knapsack 23/06/23 (B)

$$n=5 \quad p'=(15, 13, 9, 8, 3) \quad w'=(8, 6, 8, 5, 2) \quad C=18$$

Branch & Bound + numerare nodi per ordine + esplicit. upper bound se è stato necessario (calcolarlo (soltanto se necessario))

1) Ordino el. in ord. decrescente per unità di peso ($\frac{p_1}{w_1} \geq \frac{p_2}{w_2} \geq \dots \geq \frac{p_n}{w_n}$)

$$\Rightarrow \frac{15}{8} = 1,875 \quad \frac{13}{6} = 2,166 \quad \frac{9}{8} = 1,125 \quad \frac{8}{5} = 1,6 \quad \frac{3}{2} = 1,5$$

$$\Rightarrow p'=(13, 15, 8, 3, 9) \quad w'=(6, 8, 5, 2, 8)$$

$$S = \min \{i : \sum_{j=1}^i w_j > C\}; \quad \bar{C} := C - \sum_{j=1}^{S-1} w_j \quad U = \left\lfloor \sum_{j=1}^{S-1} p_j + \bar{C} \cdot \frac{p_S}{w_S} \right\rfloor$$

$$S=3 \quad \bar{C} = 18 - 6 - 8 = 4 \Rightarrow S=3$$

$$U(\text{nodo } 0) = \left\lfloor 13 + 15 + 4 \cdot \frac{8}{5} \right\rfloor = 34$$

Genero nodi 1 e 2, stesso UB di N0
 $x_1=1, x_2=1$

Nodo 3:

$$x_3=0 \text{ (no suff. cap residuo)} \Rightarrow S=5$$

$$U(N_3) = \left\lfloor 13 + 15 + 3 + 2 \cdot \frac{9}{8} \right\rfloor = 33$$

$$U(N_4) = U(N_3) = 33 \quad S=5$$

$$x_5=0 \Rightarrow x'=(1, 1, 0, 1, 0)$$

$$Z = p_1' + p_2' + p_4' = 13 + 15 + 3 = 31$$

$$N_6: x_4=0, S=5$$

$$U(N_6) = \left\lfloor 13 + 15 + 4 \cdot \frac{9}{8} \right\rfloor = 32$$

$$N_7: x_2'=0 \Rightarrow Z_1 = p_1' + p_2' = 28 < Z \Rightarrow \text{escludo } N_7$$

$$N_8: x_2=0 \Rightarrow U(N_8) = \left\lfloor 13 + 8 + 3 + (18 - 6 - 5 - 2) \cdot \frac{9}{8} \right\rfloor = 29 < Z \quad S=5$$

$$N_9: x_1=0 \Rightarrow U(N_9) = \left\lfloor 15 + 8 + 3 + (18 - 8 - 5 - 2) \cdot \frac{9}{8} \right\rfloor = 29 < Z \quad S=5$$

$\Rightarrow$ sol ottimale pari a 31

$$\text{sol intera: } x'=(1, 1, 0, 1, 0)$$

